Julian Ritondale

+54 11-2160-4006 | julianritondale@gmail.com | linkedin.com/in/julianritondale | julian-ritondale.github.io

Education

Universidad Austral — **Feb. 2023 – Dec. 2027 (Expected)**
Bachelor's degree in Software Engineering, Minor in Artificial Intelligence — *Buenos Aires, Argentina*

Relevant Coursework
Algorithms and Data Structures, Computer Architecture, Operating Systems, Databases, Artificial Intelligence, Programming Languages, Concurrent Programming, Computer Networking, Computer Security

Projects

Agent Orchestra | *Python, OpenAI API, ChromaDB, Langfuse* — **Jul. 2026**
- Collaborated building a multi-agent coding assistant orchestrating five specialized sub-agents with file, command, and search tools
- Designed a three-tier memory system to persist findings and resume interrupted tasks across sessions
- Implemented a RAG pipeline with ChromaDB and OpenAI embeddings to ground agent responses in official documentation
- Designed a permissions layer enforcing per-agent tool access, path restrictions, and command allow/deny lists
- Integrated Langfuse for observability, tracing per-agent spans, token usage/cost, and RAG retrieval quality

Snippet Searcher | *Kotlin, TypeScript, Spring Boot, React, PostgreSQL, Docker, Azure* — **Oct. 2025 – Dec. 2025**
- Collaborated building a microservices web application for managing and sharing code snippets
- Secured the platform by implementing authentication with Auth0 and request authorization through Spring Gateway
- Improved responsiveness by offloading time-intensive tasks to asynchronous inter-service communication using Redis
- Monitored system health by setting up observability in New Relic with custom attributes, dashboards, and alerts
- Automated deployments and integration checks by establishing GitHub Actions CI/CD pipelines targeting Azure VMs
- Validated critical user flows by implementing E2E tests using Cypress

PrintScript | *Kotlin, Clikt* — **Aug. 2025 – Nov. 2025**
- Collaborated developing a public library of language tools for a TypeScript subset by implementing an interpreter, formatter, and linter
- Streamlined tool usage by building a CLI for language tool execution using the Clikt library
- Enforced module independence and reusable build logic by configuring a multi-project Gradle build with shared conventions for linting, formatting, and code coverage
- Automated versioned package publishing and codebase integration by setting up GitHub Actions CI/CD pipelines deploying to Apache Maven registry
- Successfully verified library compliance through a TCK suite covering memory and runtime constraints

Sonora | *TypeScript, Tailwind CSS, React, Nest JS, PostgreSQL* — **Aug. 2025 – Nov. 2025**
- Collaborated developing a web application for browsing and reviewing music content
- Developed music discovery features by integrating the Spotify API to import artists, albums, and tracks with metadata and audio previews
- Improved API discoverability by documenting all endpoints with Swagger and OpenAPI annotations
- Enabled user engagement by implementing social features including follows, reviews, reactions, and a listen later queue

Chess and Checkers Engine | *Java* — **Jun. 2025 – Jul. 2025**
- Designed a reusable game engine using an Entity Component System architecture to share core logic across chess, checkers, and their variants
- Validated correctness of 33+ game rules by implementing a dynamic test runner and integrating a third-party test suite covering movements, captures, check, castling, and checkmate
- Enabled a playable chess and checkers UI by integrating the game engine to a provided JavaFX interface

Extracurriculars

Your Time, Your World Cup (AI Competition) | 1st Place — **Jun. 2026**
- Collaborated developing a web application that classified all 72 FIFA World Cup 2026 group-stage matches into three interest levels through user preferences and schedule availability
- Eliminated the need for historical user data by designing a Bayesian elicitation system that estimates individual utility functions through adaptive questionnaires
- Improved recommendation accuracy over time by implementing a Kalman filter that attributes user feedback to individual match features
- Reduced onboarding friction by processing natural language descriptions with the OpenAI API to automatically derive Bayesian priors and generate personalized recommendation explanations
- Simplified match planning by enabling users to add recommended matches to their calendars using the Google Calendar API

Technical Skills

Programming Languages: Kotlin, Java, TypeScript, JavaScript, HTML/CSS, Python, SQL
Libraries/Frameworks: Spring Boot, Nest JS, React, Next.js
Databases: PostgreSQL
Developer Tools: Git/GitHub, Docker, Codex, Cursor
Languages: Spanish (Native), English (Fluent/Professional)